

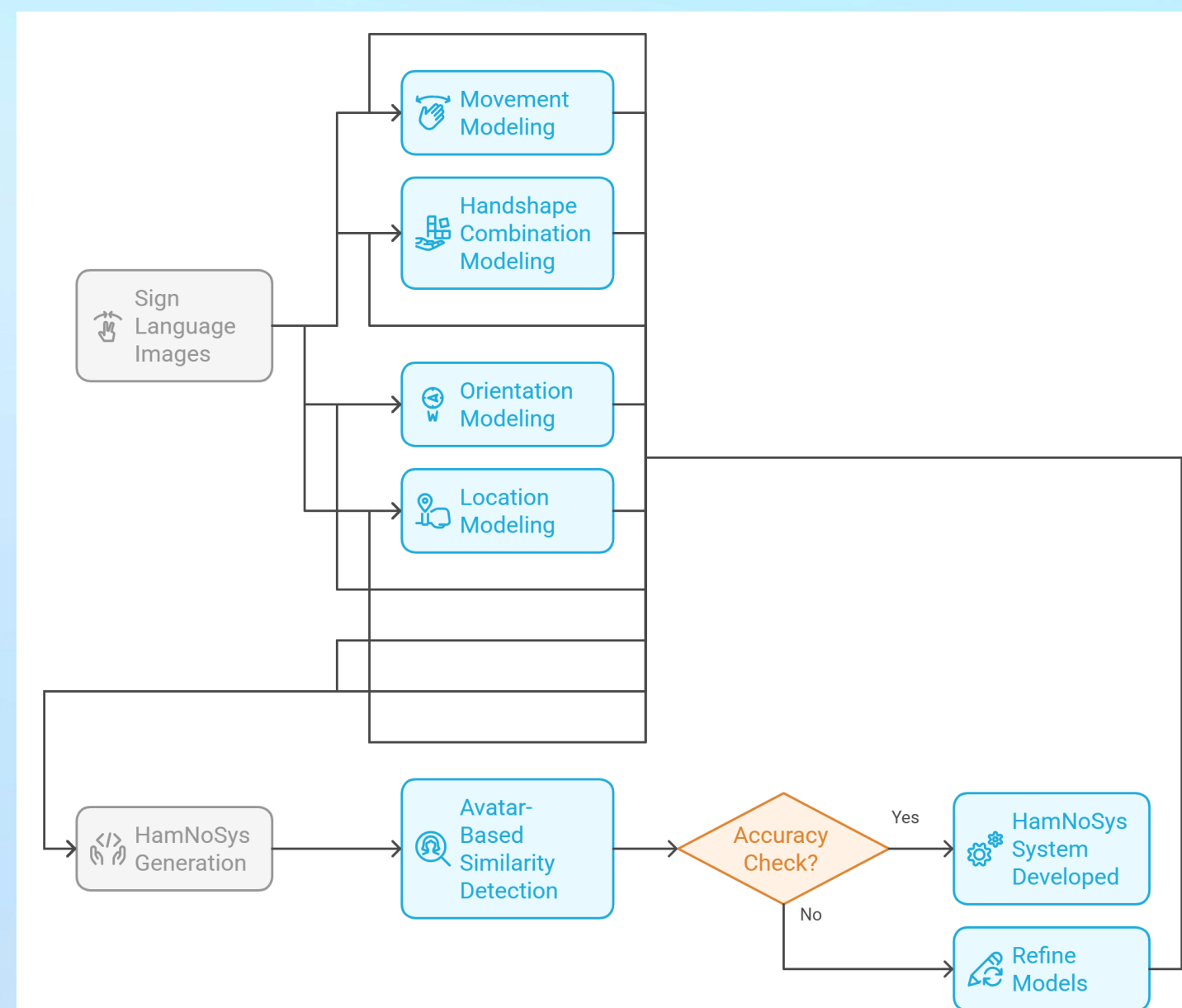
T14: Automatic HamNoSys Generation from Sign Language Videos

Objective

To automatically generate HamNoSys notation from sign language videos by analyzing handshape, orientation, location, and movement, supporting accessible sign language technology for deaf and hard-of-hearing communities.

Implementation

1. Provide sign language video as input.
2. Extract hand & body landmarks.
3. Analyze hand and body landmark positions.
4. Detect handshape, location, orientation & movement.
5. Generate HamNoSys notation.



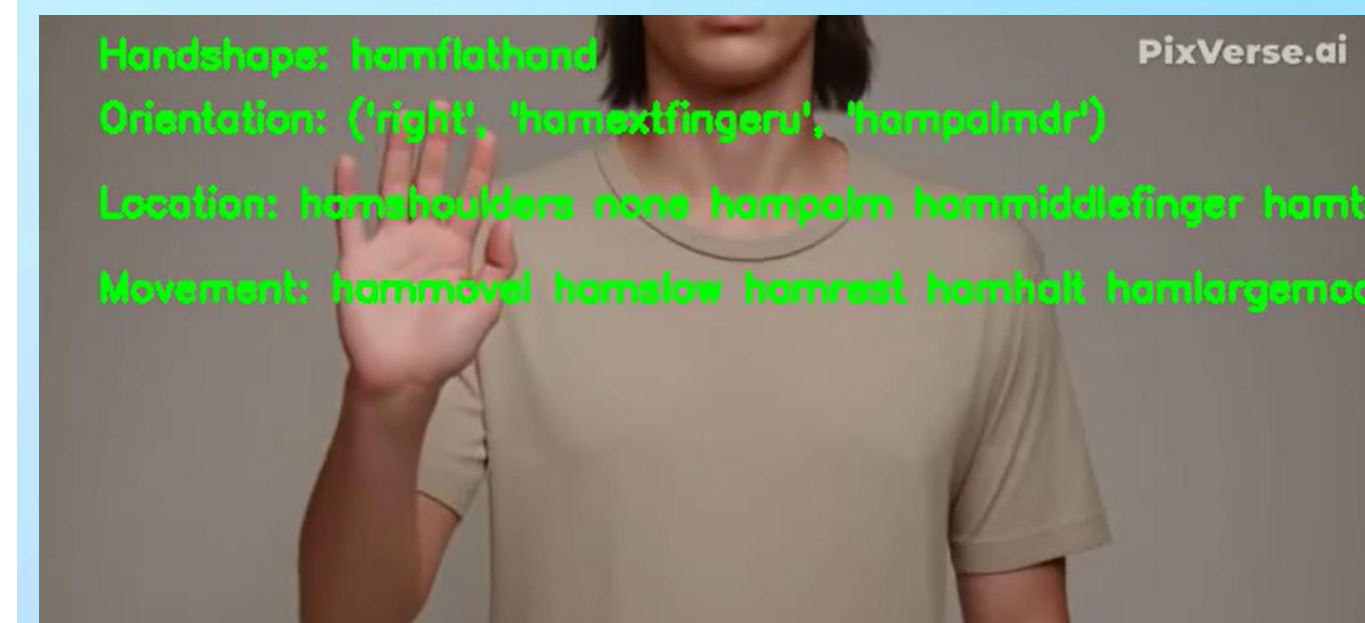
Contribution

- Custom HamNoSys dataset creation.
- ML-based handshape recognition.
- Rule-based orientation, location & movement detection using mathematical formulations.
- Anatomically justified geometric thresholds.

Guide

Ms. Nalinadevi K
Assistant Professor

Results



The input sign language video is processed and divided into frames for analysis. Each frame is used to detect handshape, orientation, location, and movement to generate the corresponding HamNoSys notation.

Conclusion

The system successfully generates HamNoSys representations from sign language videos using landmark-based analysis, eliminating manual annotation while maintaining consistent results.

Team

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